Kyle Rebello

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Education

McMaster University — Hamilton, ON

Bachelor of Engineering in Computer Engineering (Co-op) — *Expected Graduation: April 2029*

- **Awards**: Dean's Honour List (2025), Faculty of Engineering Award of Excellence (2024)
- **Relevant Coursework**: Digital Systems Design, Digital Logic Design, Algorithm Design and Analysis, Electronic Circuits and Devices II, Linear Algebra, Calculus III, Advanced Probability and Random Processes, Signals and Systems

Work Experience

ACTO — May 2026 – Present

AI Engineer Intern — *Toronto, ON*

- Cut median quote-build time **4×** (from **∼120 to 30 min**) for **10+ sales reps**, by replacing a legacy Excel pricing calculator with a natural-language LLM sales agent
- Ensured every quoted price is computed deterministically, by architecting a hybrid system in which a deterministic pricing engine calculates all figures and the LLM only narrates them, confining hallucination and prompt-injection risk to explanatory text
- Delivered citation-backed answers traceable to source documents, by building a retrieval-augmented generation (RAG) pipeline over an Amazon OpenSearch Serverless vector index
- Shipped the full-stack B2B platform hosting the agent, including SSO, three-tier role-based access control, PDF order-form generation, and token-secured proposal pages with engagement analytics

Weever — Apr. 2026 – Present

Machine Learning Engineer (Part-time Contract) — *Hamilton, ON*

- Enabled **zero-configuration** onboarding of any customer's operational logs, by building a schema-agnostic Python pipeline (pandas, SciPy) that infers column types from content rather than names, with seeded, reproducible scoring
- Segmented workforces into behavioural archetypes, by engineering **45 features** (submission cadence, evidence quality, measurement consistency) and selecting between k-means and Gaussian mixture models using silhouette scores in scikit-learn
- Gave managers a ranked watchlist of workers showing engagement drift, by building a composite risk score that compares each worker's output, documentation quality, and work patterns against their own baseline and their team's trend
- Let managers query results in plain English, by shipping the analytics layer as a self-contained local app with an MCP server for LLM-based natural-language querying

Projects

INT-8 FPGA Neural Network Accelerator | *Verilog, PyTorch, Python, Quartus* — Sept. 2026 – Oct. 2026

- Built a hardware neural network accelerator that classifies handwritten digits in **130 μs** at **97%** accuracy, by designing a 16-unit parallel multiply-accumulate (MAC) engine in Verilog on a Cyclone IV FPGA (DE2-115)
- Cut model memory **4×** with under **0.5%** accuracy loss, by quantizing a PyTorch-trained network from 32-bit floats to 8-bit integers using hardware-friendly power-of-2 rescaling
- Identified data transfer as the main bottleneck (**500×** slower than compute) and mapped speed vs. chip-area trade-offs across 8/16/32-MAC designs, by profiling with on-chip cycle counters, Quartus timing reports, and a logic analyzer
- Achieved a **100%** bit-exact match with a Python golden model across all **10,000 MNIST test images**, by building self-checking Verilog testbenches and an automated Python test harness over UART

Edge AI Optimization on Jetson Nano | *C++, CUDA, TensorRT, ONNX, PyTorch* — Sept. 2026 – Oct. 2026

- Cut ResNet-18 inference latency **3×** (from **34 ms** in PyTorch to **11 ms**) with under **0.5%** accuracy change, by exporting the model to ONNX and building an FP16 inference pipeline in C++ with TensorRT's C++ API
- Sped up matrix multiplication **4×** over a naive CUDA kernel, reaching **30%** of cuBLAS performance, by writing shared-memory tiled kernels in CUDA C++ and profiling them with nvprof, Nsight Systems, and a roofline model
- Measured **60 mJ** per inference and compared performance-per-watt across 5W and 10W power modes, by building a shunt-resistor measurement rig with an Analog Discovery 3 and validating it against the Jetson's onboard power sensors

Technical Skills

Languages: Python, C/C++, Verilog/SystemVerilog, SQL (PostgreSQL, SQLite), MATLAB
Machine Learning: PyTorch, scikit-learn, pandas, NumPy, SciPy, LLMs, RAG, vector databases, INT8 quantization
Cloud & Dev Tools: AWS (Bedrock, OpenSearch Serverless, ECS, ECR, RDS, S3), Docker, Linux, Git, CI/CD
GPU & ML Acceleration: CUDA, cuBLAS, TensorRT, ONNX, Nsight Systems, nvprof, roofline analysis
Hardware: FPGA design (Cyclone IV), Quartus, ModelSim, NVIDIA Jetson Nano
Analog & Digital Electronics: Operational Amplifiers, Multistage Amplifier, Oscillators, Field-effect and Bipolar Transistors, Circuit Design & Analysis (Analog Discovery 3, PSpice)

Extracurriculars

McMaster Dragon Boat Competitive Rowing Team, Men's Intramural Soccer